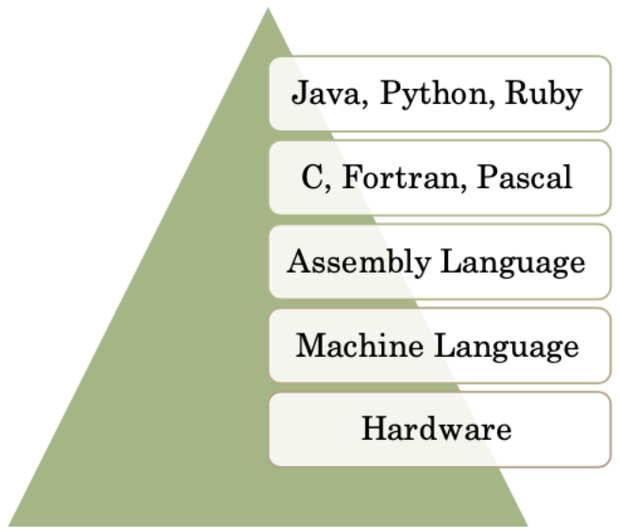
CSE307
Principles of Programming Languages

State University of New York, Korea

Course homepage

Programming Languages Evolution

- Most programming languages are *high-level* languages, where the phrase "high-level" indicates a higher degree of abstraction.
- The second word of this course's name – **abstraction** – is among the most important ideas in programming.
- It refers to the degree to which the language's features are separated away from the details of a particular computer's architecture and/or implementation at *lower* levels.



Abstraction

- A round, fluffy cat of 15 kilos with light orange and white fur
- A round, fluffy cat of 15 kilos
- A round cat
- Cat
- Animal

This course

- Syntax and grammar
- Lambda calculus
- Understanding programming languages via OCaml and a bit of C

Logistics

Meet the Instructor

- Man email: <zhoulai dot fu at sunykorea dot ac dot kr>
- Affiliated faculty with CS, SBU and ECE, VT
- CSE113 and CSE307
- Research Interest: Programming Language Theory, Software Security, AI systems; floating-point computation
- Previous Work: France, Spain, US, Denmark
- Education: École Polytechnique, France
- Personal: Happily married; no special hobbies

Course website

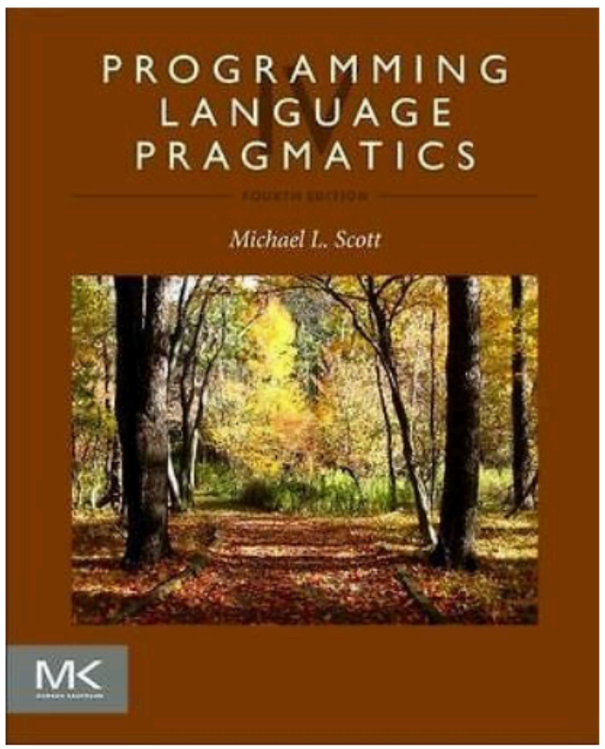
https://github.com/zhoulaifu/26_cse307_fall

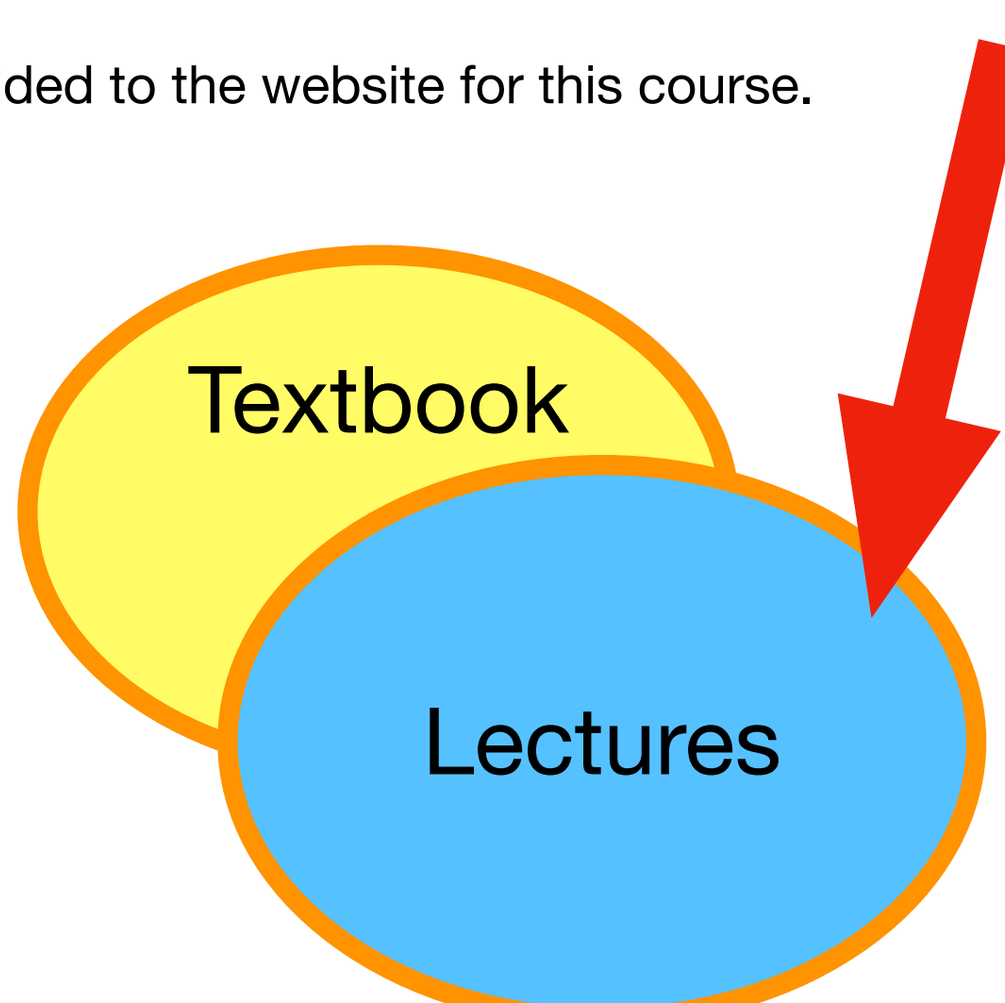
Reference books and reading material

- Michael L. Scott. Programming Language Pragmatics.

- For details pertaining to specific programming languages, the recommended material will mostly be from the following:
  Python tutorial: https://docs.python.org/3/tutorial/
  The official OCaml learning material from https://ocaml.org/learn/

- Other reading material (if used) will be added to the website for this course.





Questions so far?

Quiz

- Where to find official course info?
- Homework due time?
- How attendance will be checked?
- How late homework will be handled?
- How to email the instructor?
- How grading works?

Programming Languages Paradigms

Programming Paradigms

- What: A style or approach to programming that provides **a framework for building a software system.** It is a **set of principles, concepts, and practices** that define the way of thinking and organizing the code to solve a specific problem.

- Main paradigms: Imperative, procedural, reactive, declarative, object-oriented, functional

- Demo

- **Imperative Programming:** This paradigm focuses on the sequence of **statements that modify the state of the program**, and how to control that sequence using conditional statements, loops, and other control flow constructs.

- **Object-Oriented Programming (OOP)**: This paradigm is based on the idea of **organizing software systems as a collection of objects** that interact with each other to solve a problem.

- **Functional Programming:** This paradigm emphasizes **the use of functions** to solve problems and avoid changing the state of the program.

- **Declarative Programming:** This paradigm focuses on **describing the problem** to be solved, rather than how to solve it.

Declarative programming languages

- In declarative programming, the program **describes the desired result, rather than specifying how to achieve it.** This is in contrast to procedural programming, where the program specifies a series of steps to accomplish a task.

- One example of declarative programming is SQL (Structured Query Language), which is used to query relational databases. In SQL, a programmer specifies the criteria for selecting records from a database, and the database management system determines how to retrieve the data.

- Another example is HTML. Programmers specify web structure and content using tags, and the web browser determines how to render the page based on those tags.

Declarative programming in SQL

```sql
-- Define a table of students
CREATE TABLE students (
  id INT,
  name VARCHAR(255),
  major VARCHAR(255),
  gpa FLOAT
);

-- Select all students with a GPA greater than 3.0
SELECT id, name, major
FROM students
WHERE gpa > 3.0;
```

- Note that we're not specifying how to retrieve the data or iterating over it step-by-step, but rather describing the desired result in a declarative way. SQL takes care of the details of how to retrieve the data and return the desired result.

- So, this is declarative programming - we declare what we want to happen, and the programming language takes care of the details.

Declarative programming in HTML

```html
<!DOCTYPE html>
<html>
  <head>
    <title>My Web Page</title>
  </head>
  <body>
    <h1>Welcome to My Web Page</h1>
    <p>This is a paragraph of text.</p>
    <ul>
      <li>Item 1</li>
      <li>Item 2</li>
      <li>Item 3</li>
    </ul>
  </body>
</html>
```

- Note that we're not specifying how the page should be rendered, but rather describing the structure and content of the page in a declarative way. The web browser takes care of the details of how to render the page.

Functional Programming

1

Based on recursive definitions.

- They are inspired by a computational model called **lambda calculus**, developed by Alonzo Church in the 1930s.

2

A program is viewed as a mathematical function that transforms an input to an output. It is often defined in terms of simpler functions.

- We will see many examples of functional programming in multiple languages (e.g., Java, Python, OCaML).

Functional Programming in Python

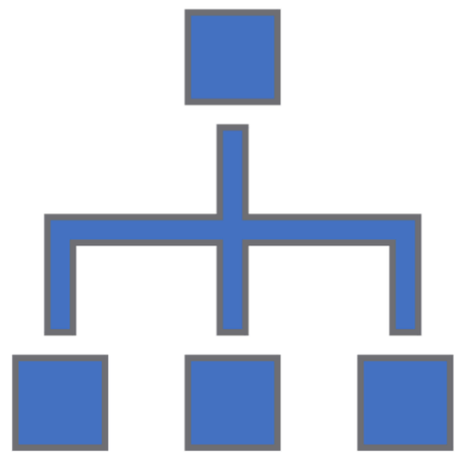
```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]


even_numbers = filter(lambda x: x % 2 == 0, numbers)
squared_numbers = map(lambda x: x ** 2, even_numbers)
sum_of_squares = sum(squared_numbers)
```

Functional Programming in Java

```
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6, 7, 8, 9, 10);

int sumOfSquares = numbers.stream()
    .filter(x -> x % 2 == 0)
    .map(x -> x * x)
    .reduce(0, Integer::sum);
```

Functional Programming in Ocaml

```
let numbers = [1; 2; 3; 4; 5; 6; 7; 8; 9; 10];;


let is_even x = x mod 2 = 0;;

let sum_of_squares =
  List.filter is_even numbers
  |> List.map (fun x -> x * x)
  |> List.fold_left (+) 0;;
```



- One more example. Three paradigms.

- Implementing the greatest common divisor (GCD) solution in different paradigms and different languages.

The GCD problem: pseudocode

```
function gcd(a, b)
    while a ≠ b
        if a > b
            a := a − b;
        else
            b := b − a;
    return a;
```

Paradigm 1:
Imperative Programming in Python

```
1  def gcd (x, y):
2      while (x!=y):
3          if (x>y): x = x-y
4          else: y = y -x
5      return x
6
7  x = 15
8  y = 40
9  print (gcd(x,y))
```

Paradigm 2:
Object-Oriented Programming in Java

/MyClass.java

```java
public class MyClass {



    static int gcd (int x, int y){
        while (x!=y){
            if (x>y) x = x-y;
            else y = y -x;
        }
        return x;
    }



    public static void main(String args[]) {
      int x=10;
      int y=25;


      System.out.println("GCD of x+y = "+ gcd(x,y));
    }
}
```

Paradigm 3:
Functional Programming in Ocaml

```
1  let rec gcd (x, y) = if x> y then gcd (x-y, y) else if x < y then gcd (x, y -x) else x ;;
2
3  let r= gcd(15, 40);;
4
5  print_int r;;
6
```

Why Harvard, MIT, Stanford, Cambridge all teach functional programming

https://youtu.be/6APBx0WsgeQ

Summary

- Imperative, functional, object-oriented paradigms

- A significant part of understanding programming is about understanding programming language paradigms

- We will focus on functional programming soon